

HCSM-27

Klein Orbit Structure

The Klein Four Group Action, Kernel Classification, and the Framework Constant Four Orbit
The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self paired eigenvalue of the discrete Laplacian on the periodic eight by eight torus to one Standard Model generation. Two order two operations act naturally on this multiplet: the diagonal reflection, identified in HCSM-26 as the unique chiral reflection, and the half-period translation, the diagonal shift of both mode coordinates by half the torus side length. The two operations commute, and each squares to the identity, so they generate a Klein four group. This paper proves three theorems about the action of this Klein group on the fourteen mode multiplet. First, the action has orbit decomposition five size two orbits plus one size four orbit, totalling fourteen modes. Second, the pointwise stabilizer of each orbit falls into one of three classes, corresponding to the three non trivial subgroups of the Klein group. Third, the four cycle orbit has four mass-ordering values that are simultaneously all framework constants: nine, twelve, fourteen, and twenty one. These values satisfy two additive identities within the orbit, and their sum equals fifty six, the dimension of the Hodge sector at the self paired eigenvalue; their mean equals fourteen, the multiplicity. This paper also proves that the orbit class swap pair, consisting of the dark photon and the top quark, is the unique edge in the distinguished pair graph that connects two distinct Klein orbits. All theorems are HCSM native; all numerical claims are verified at machine precision.

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1 Introduction

1.1 The Klein group in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight by eight torus [1]. The fourteen modes at the self paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation. Their dihedral representation theory determines the gauge algebra; their mass ordering labels determine the mass spectrum; the zero state vector from the fourteen by nineteen measurement matrix determines the flavor structure.

Two order two operations act on the fourteen mode multiplet. The first is the diagonal reflection, which swaps the two mode coordinates and has been identified in HCSM-26 as the unique chiral reflection of the dihedral group [6]. The second is the half period translation, which shifts both mode coordinates by half the torus side length. The two operations commute, and each squares to the identity, so they generate a Klein four group.

This paper proves that the action of the Klein group organizes the fourteen mode multiplet completely: the orbits are classified, the stabilizers of the orbits are classified, and one orbit in particular the four cycle orbit contains four mass ordering values that are all framework constants.

1.2 The one line summary

The Klein group is the framework's smallest non trivial symmetry group that acts on the fourteen mode multiplet in a way that is neither dihedral nor trivial. Its orbit structure separates the fourteen modes into three regions: four two cycle orbits with a common stabilizer, an observer orbit fixed by the chiral reflection, and a four cycle orbit with trivial stabilizer. The four cycle orbit is the framework-constant orbit.

1.3 The three theorems of this paper

The paper is organized around three theorems.

1. **Klein Orbit Theorem** (Section 3). The Klein four group generated by the diagonal reflection and the half period translation acts on the fourteen mode multiplet with orbit decomposition consisting of five size two orbits and one size four orbit.
2. **Kernel Classification Theorem** (Section 4). The pointwise stabilizer of each Klein orbit falls into one of three classes, corresponding to the three non trivial subgroups of the Klein group.
3. **Four Orbit Framework Constant Theorem** (Section 6). The four cycle orbit's four mass ordering values are all framework constants, and their sum equals fifty six, the dimension of the Hodge sector at the self paired eigenvalue.

Section 5 proves a companion theorem the Swap Pair Bridge Theorem identifying the unique inter orbit edge in the distinguished-pair graph.

1.4 Cross references to the HCSM paper stack

The paper depends on two companion papers in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-14** [3]: the winding generator spectrum and the class partition. Used for the class assignments.
- **HCSM-26** [6]: the observer pair. Used for the diagonal reflection, the observer pair definition, and the fixed-locus theorem.

Additional references to HCSM-13 (the processor T), HCSM-22 (the zero state), and HCSM-23 (the mass-ordering labels) appear where needed.

1.5 Organization

Section 2 states the setup: the fourteen mode multiplet, the diagonal reflection, the half period translation, and the Klein structure. Section 3 proves the Klein Orbit Theorem. Section 4 proves the Kernel Classification Theorem. Section 5 proves the Swap Pair Bridge Theorem. Section 6 proves the Four Orbit Framework Constant Theorem. Section 7 discusses the structural interpretation. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup

This section states the HCSM primitives on which the paper depends. All statements are HCSM-native; the reader is referred to HCSM-00 for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight by eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1]. The discrete five point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self paired eigenvalue $\lambda = 4$ has multiplicity fourteen. In the standard mode order used throughout HCSM (matching the zero state vector of HCSM-22 [4]), the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (2)$$

The mode coordinates in this order are:

$$\begin{aligned} \nu : (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \quad s : (3, 1), \quad \mu : (3, 7), \\ \text{dark} : (5, 7), \quad c : (7, 3), \quad \tau : (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H : (6, 2), \quad t : (6, 6). \end{aligned} \quad (3)$$

2.2 The diagonal reflection

The diagonal reflection is the map

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (4)$$

It is an element of the dihedral point group of the square, and it satisfies $rs^2 = e$. It has been identified in HCSM-26 as the unique chiral reflection of the dihedral group: the unique element that fixes exactly the pair $\{(2, 2), (6, 6)\}$ on the fourteen mode multiplet and anticommutes with the winding generator.

The fixed set of rs on the fourteen mode multiplet is exactly the observer pair $\{W, t\}$ [6].

2.3 The half-period translation

The half-period translation is the map

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4, n_2 + 4) \pmod{8}. \quad (5)$$

It shifts both mode coordinates by half the torus side length $L/2 = 4$. It satisfies $T_{\text{half}}^2 = e$.

2.4 The Klein four-group

Lemma 2.1 (Klein structure). *The diagonal reflection rs and the half-period translation T_{half} commute and each squares to the identity:*

$$rs^2 = e, \quad T_{\text{half}}^2 = e, \quad [rs, T_{\text{half}}] = 0. \quad (6)$$

Hence they generate a Klein four-group

$$V = \langle rs, T_{\text{half}} \rangle = \{e, rs, T_{\text{half}}, rs \cdot T_{\text{half}}\}. \quad (7)$$

Proof. The relations $rs^2 = e$ and $T_{\text{half}}^2 = e$ are immediate from the definitions. For the commutator: for any mode (n_1, n_2) ,

$$\begin{aligned} rs(T_{\text{half}}(n_1, n_2)) &= rs(n_1 + 4, n_2 + 4) = (n_2 + 4, n_1 + 4), \\ T_{\text{half}}(rs(n_1, n_2)) &= T_{\text{half}}(n_2, n_1) = (n_2 + 4, n_1 + 4). \end{aligned}$$

The two expressions coincide. Hence $[rs, T_{\text{half}}] = 0$. The group generated by two commuting involutions is the Klein four group; this is verified at machine precision in the companion verification script. $\square$

2.5 The product $rs \cdot T_{\text{half}}$

The product of the two generators is the map

$$rs \cdot T_{\text{half}} : (n_1, n_2) \mapsto (n_2 + 4, n_1 + 4) \pmod{8}. \quad (8)$$

It is an involution, and it is not equal to either rs or T_{half} .

2.6 The mass-ordering labels

The mass ordering labels (the $N/4$ values) are derived from the mode coordinates by the δ function of HCSM-23 [5]. In the standard mode order,

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (9)$$

Throughout this paper, we refer to these values using descriptive names, e.g., the dark-photon label equals twelve, and the top-quark label equals one half.

3 The Klein Orbit Theorem

This section proves that the Klein group acts on the fourteen mode multiplet with five size two orbits and one size four orbit.

3.1 Orbit decomposition

Theorem 3.1 (Klein Orbit Theorem). *The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ acts on the fourteen mode multiplet with orbit decomposition consisting of five orbits of size two and one orbit of size four. Explicitly, the orbits are:*

Orbit	Members	Size
V_1	$\{\nu, e\}$	2
V_2	$\{H, Z\}$	2
V_3	$\{d, b\}$	2
V_4	$\{\mu, c\}$	2
V_5	$\{W, t\}$	2
V_6	$\{\text{dark}, s, u, \tau\}$	4

The total mode count is $5 \cdot 2 + 1 \cdot 4 = 14$, matching the multiplicity at the self-paired eigenvalue.

Proof. Direct computation of the orbit of each mode under the Klein group. We compute representative orbits.

Orbit of ν at $(0, 4)$:

$$\begin{aligned} e \cdot \nu &= (0, 4) = \nu, \\ rs \cdot \nu &= (4, 0) = e, \\ T_{\text{half}} \cdot \nu &= (4, 0) = e, \\ (rs \cdot T_{\text{half}}) \cdot \nu &= (0, 4) = \nu. \end{aligned}$$

The orbit is $\{\nu, e\}$, size 2. This is V_1 .

Orbit of H at $(6, 2)$:

$$\begin{aligned} e \cdot H &= (6, 2) = H, \\ rs \cdot H &= (2, 6) = Z, \\ T_{\text{half}} \cdot H &= (2, 6) = Z, \\ (rs \cdot T_{\text{half}}) \cdot H &= (6, 2) = H. \end{aligned}$$

The orbit is $\{H, Z\}$, size 2. This is V_2 .

Orbit of d at $(1, 5)$:

$$\begin{aligned} e \cdot d &= (1, 5) = d, \\ rs \cdot d &= (5, 1) = b, \\ T_{\text{half}} \cdot d &= (5, 1) = b, \\ (rs \cdot T_{\text{half}}) \cdot d &= (1, 5) = d. \end{aligned}$$

The orbit is $\{d, b\}$, size 2. This is V_3 .

Orbit of μ at $(3, 7)$:

$$\begin{aligned} e \cdot \mu &= (3, 7) = \mu, \\ rs \cdot \mu &= (7, 3) = c, \\ T_{\text{half}} \cdot \mu &= (7, 3) = c, \\ (rs \cdot T_{\text{half}}) \cdot \mu &= (3, 7) = \mu. \end{aligned}$$

The orbit is $\{\mu, c\}$, size 2. This is V_4 .

Orbit of W at $(2, 2)$:

$$\begin{aligned} e \cdot W &= (2, 2) = W, \\ rs \cdot W &= (2, 2) = W, \\ T_{\text{half}} \cdot W &= (6, 6) = t, \\ (rs \cdot T_{\text{half}}) \cdot W &= (6, 6) = t. \end{aligned}$$

The orbit is $\{W, t\}$, size 2. This is V_5 .

Orbit of dark at $(5, 7)$:

$$\begin{aligned} e \cdot \text{dark} &= (5, 7) = \text{dark}, \\ rs \cdot \text{dark} &= (7, 5) = \tau, \\ T_{\text{half}} \cdot \text{dark} &= (1, 3) = u, \\ (rs \cdot T_{\text{half}}) \cdot \text{dark} &= (3, 1) = s. \end{aligned}$$

The orbit is $\{\text{dark}, \tau, u, s\}$, size 4. This is V_6 .

The six orbits cover all fourteen modes. The orbit decomposition is as in the statement. $\square$

3.2 The three structural regions

The orbit decomposition partitions the fourteen-mode multiplet into three structural regions:

The four two-cycle orbits V_1 – V_4 .

Each of these orbits has two members that are exchanged by the diagonal reflection and by the half-period translation, and that are fixed by the product $rs \cdot T_{\text{half}}$. They are the framework's standard mirror pairs.

The observer orbit V_5 .

This orbit is $\{W, t\}$, the observer pair of HCSM-26. Both members are fixed by the diagonal reflection rs .

The four cycle orbit V_6 .

This orbit contains four modes that are cyclically exchanged by the Klein group. It contains the swap partner of the top quark (see Section 5) and has the framework-constant property of Section 6.

Remark 3.2. The four-cycle orbit V_6 contains the modes u, s, dark, τ , all drawn from the middle dihedral orbit $\mathcal{O}_1$. The Klein group therefore further refines $\mathcal{O}_1$: the dihedral orbit of size 8 splits into the three Klein orbits $V_3 = \{d, b\}$, $V_4 = \{\mu, c\}$, and $V_6 = \{\text{dark}, s, u, \tau\}$.

4 Kernel Classification

This section classifies the pointwise stabilizer of each Klein orbit. The stabilizer of an orbit is the set of group elements that fix every member of the orbit.

4.1 Definition

Definition 4.1 (Pointwise stabilizer). For a Klein orbit $V_i \subset \mathcal{M}$, the *pointwise stabilizer* is the subgroup of V consisting of elements g such that $g \cdot m = m$ for every mode $m \in V_i$.

4.2 Classification

Theorem 4.2 (Kernel Classification Theorem). *The pointwise stabilizer of each Klein orbit is one of the three non-trivial subgroups of the Klein group. The classification is:*

<i>Orbit</i>	<i>Members</i>	<i>Pointwise stabilizer</i>
V_1	$\{\nu, e\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_2	$\{H, Z\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_3	$\{d, b\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_4	$\{\mu, c\}$	$\{e, rs \cdot T_{\text{half}}\}$
V_5	$\{W, t\}$	$\{e, rs\}$
V_6	$\{\text{dark}, s, u, \tau\}$	$\{e\}$

Proof. Direct computation of the action of each group element on each orbit. We compute representative cases.

Orbit $V_1 = \{\nu, e\}$. The mode ν is at $(0, 4)$ and the mode e is at $(4, 0)$.

$$\begin{aligned}
 rs \cdot \nu &= (4, 0) = e \neq \nu \quad (\text{swaps}), \\
 T_{\text{half}} \cdot \nu &= (4, 0) = e \neq \nu \quad (\text{swaps}), \\
 (rs \cdot T_{\text{half}}) \cdot \nu &= (0, 4) = \nu, \quad (rs \cdot T_{\text{half}}) \cdot e = (4, 0) = e \quad (\text{fixed pointwise}).
 \end{aligned}$$

Hence the pointwise stabilizer is $\{e, rs \cdot T_{\text{half}}\}$.

Orbit $V_5 = \{W, t\}$. The mode W is at $(2, 2)$ and the mode t is at $(6, 6)$.

$$\begin{aligned}
 rs \cdot W &= (2, 2) = W, \quad rs \cdot t = (6, 6) = t \quad (\text{fixed pointwise}), \\
 T_{\text{half}} \cdot W &= (6, 6) = t \neq W \quad (\text{swaps}), \\
 (rs \cdot T_{\text{half}}) \cdot W &= (6, 6) = t \neq W \quad (\text{swaps}).
 \end{aligned}$$

Hence the pointwise stabilizer is $\{e, rs\}$.

Orbit $V_6 = \{\text{dark}, s, u, \tau\}$. The mode dark is at $(5, 7)$.

$$\begin{aligned}
 rs \cdot \text{dark} &= (7, 5) = \tau \neq \text{dark} \quad (\text{swaps}), \\
 T_{\text{half}} \cdot \text{dark} &= (1, 3) = u \neq \text{dark} \quad (\text{swaps}), \\
 (rs \cdot T_{\text{half}}) \cdot \text{dark} &= (3, 1) = s \neq \text{dark} \quad (\text{swaps}).
 \end{aligned}$$

No non-identity element fixes every member of V_6 . Hence the pointwise stabilizer is $\{e\}$.

The remaining orbits are computed analogously and yield the classification as in the statement. $\square$

4.3 The three stabilizer classes

Corollary 4.3 (Three stabilizer classes). *The Klein orbits partition into three stabilizer classes:*

- **Class** $\{e, rs \cdot T_{\text{half}}\}$: *four orbits, V_1, V_2, V_3, V_4 .*
- **Class** $\{e, rs\}$: *one orbit, V_5 .*
- **Class** $\{e\}$: *one orbit, V_6 .*

Proof. Immediate from the theorem. □

Remark 4.4. The observer orbit V_5 is the unique orbit whose pointwise stabilizer is generated by the chiral reflection rs . This is consistent with the Chirality Selection Theorem of HCSM-26: the observer pair is the unique fixed locus of rs on the fourteen-mode multiplet.

5 The Swap-Pair Bridge Theorem

This section proves that the orbit-class swap pair is the unique edge in the distinguished-pair graph that connects two distinct Klein orbits.

5.1 The distinguished-pair graph

Definition 5.1 (Distinguished-pair graph). The *distinguished-pair graph* of the fourteen-mode multiplet has a vertex for each mode and an edge for each two-element subset that is distinguished by at least one structural criterion. The criteria are: (i) being fixed by an involution of the dihedral group; (ii) being a two-cycle of the diagonal reflection rs ; (iii) being a two-cycle of the half-period translation T_{half} ; and (iv) being the orbit-class swap pair (defined in HCSM-26).

The graph has ten edges, listed in HCSM-26 [6]. Its vertex set is a subset of the fourteen modes.

5.2 The inter-orbit edges

Theorem 5.2 (Swap-Pair Bridge Theorem). *The orbit-class swap pair $\{\text{dark}, t\}$ is the unique edge in the distinguished-pair graph that connects two distinct Klein orbits.*

Proof. The ten edges of the distinguished-pair graph are, from HCSM-26:

$$\begin{aligned} &\{\nu, e\}, \quad \{u, s\}, \quad \{u, \text{dark}\}, \quad \{d, b\}, \quad \{\mu, c\}, \\ &\{s, \tau\}, \quad \{\text{dark}, \tau\}, \quad \{\text{dark}, t\}, \quad \{Z, H\}, \quad \{W, t\}. \end{aligned}$$

We examine each edge.

Within-orbit edges. The following nine edges have both endpoints in the same Klein orbit:

$$\begin{aligned} &\{\nu, e\} \subset V_1, \\ &\{Z, H\} \subset V_2, \\ &\{d, b\} \subset V_3, \\ &\{\mu, c\} \subset V_4, \\ &\{W, t\} \subset V_5, \\ &\{u, s\}, \quad \{u, \text{dark}\}, \quad \{s, \tau\}, \quad \{\text{dark}, \tau\} \subset V_6. \end{aligned}$$

The remaining edge. The edge $\{\text{dark}, t\}$ has its two endpoints in distinct Klein orbits:

$$\text{dark} \in V_6 = \{\text{dark}, s, u, \tau\}, \quad t \in V_5 = \{W, t\}. \quad (10)$$

Hence $\{\text{dark}, t\}$ is the unique inter-orbit edge. □

Corollary 5.3 (Bridge structure). *The swap pair $\{\text{dark}, t\}$ is the unique edge of the distinguished-pair graph that bridges the four-cycle orbit V_6 and the observer orbit V_5 . It is the only connection between the framework-constant orbit and the observer orbit.*

Proof. Immediate from the theorem. $\square$

6 The Four-Orbit Framework-Constant Theorem

This section proves that the four-cycle orbit has four mass-ordering values that are all framework constants.

6.1 The four-cycle orbit's values

The four-cycle orbit is $V_6 = \{\text{dark}, s, u, \tau\}$. Its four modes have mass-ordering labels:

$$N/4(\tau) = 9, \quad (11)$$

$$N/4(\text{dark}) = 12, \quad (12)$$

$$N/4(s) = 14, \quad (13)$$

$$N/4(u) = 21. \quad (14)$$

6.2 The theorem

Theorem 6.1 (Four-Orbit Framework-Constant Theorem). *The four mass-ordering labels of the four-cycle orbit V_6 are all simple framework constants:*

$$N/4(\tau) = (d - 1)^2 = 9, \quad (15)$$

$$N/4(\text{dark}) = (d - 1) \cdot d = 12, \quad (16)$$

$$N/4(s) = \text{multiplicity} = 14, \quad (17)$$

$$N/4(u) = (d - 1)(2d - 1) = 21. \quad (18)$$

They satisfy two additive identities within the orbit:

$$N/4(\tau) + N/4(\text{dark}) = N/4(u), \quad (19)$$

$$N/4(\tau) + N/4(s) = H + 2d - 1 = 23, \quad (20)$$

where $H = 2^d = 16$ is the hidden-sector dimension parameter. The sum of the four labels equals

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = 9 + 12 + 14 + 21 = 56, \quad (21)$$

which is the dimension of the Hodge sector at the self-paired eigenvalue. The mean of the four labels equals

$$\frac{56}{4} = 14, \quad (22)$$

which is the multiplicity at the self-paired eigenvalue.

Proof. Direct arithmetic on the values. For the additive identities:

$$9 + 12 = 21 \quad (= N/4(u)), \quad (23)$$

$$9 + 14 = 23 \quad (= H + 2d - 1). \quad (24)$$

For the sum:

$$9 + 12 + 14 + 21 = 56 = 14 + 28 + 14 = \dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2)|_{\lambda=4}. \quad (25)$$

For the mean:

$$\frac{56}{4} = 14 = 2(L - 1) = \text{multiplicity}_{\lambda=4}. \quad (26)$$

$\square$

6.3 The algebraic derivation of the sum

Proposition 6.2 (Algebraic derivation of the sum). *The sum of the four-cycle orbit's mass-ordering labels has the framework-constant form*

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = 2(d-1)(2d-1) + 14. \quad (27)$$

At $d = 4$, this evaluates to $2 \cdot 3 \cdot 7 + 14 = 42 + 14 = 56$.

Proof. Add the four framework-constant forms:

$$N/4(\tau) + N/4(\text{dark}) + N/4(s) + N/4(u) = (d-1)^2 + (d-1)d + 14 + (d-1)(2d-1). \quad (28)$$

Collect the terms in $(d-1)$:

$$= (d-1)[(d-1) + d + (2d-1)] + 14 \quad (29)$$

$$= (d-1)[4d-2] + 14 \quad (30)$$

$$= 2(d-1)(2d-1) + 14. \quad (31)$$

At $d = 4$: $2 \cdot 3 \cdot 7 + 14 = 42 + 14 = 56$. $\square$

Corollary 6.3 (Four-orbit as a shadow of the Hodge sector). *The four-cycle orbit is a four-element shadow of the fifty-six-dimensional Hodge sector at the self-paired eigenvalue: its four mass-ordering labels sum to the sector dimension, and their mean is the multiplicity.*

Proof. Immediate from the theorem. $\square$

7 Discussion

7.1 What the Klein structure gives

The Klein four-group $V = \langle rs, T_{\text{half}} \rangle$ organizes the fourteen-mode multiplet into six orbits. The orbit decomposition is:

- Four two-cycle orbits with pointwise stabilizer $\{e, rs \cdot T_{\text{half}}\}$: $V_1 = \{\nu, e\}$, $V_2 = \{H, Z\}$, $V_3 = \{d, b\}$, $V_4 = \{\mu, c\}$.
- One two-cycle orbit with pointwise stabilizer $\{e, rs\}$: the observer orbit $V_5 = \{W, t\}$.
- One four-cycle orbit with trivial pointwise stabilizer: $V_6 = \{\text{dark}, s, u, \tau\}$.

The orbit-class swap pair is the unique bridge between the four-cycle orbit and the observer orbit. This is the framework's minimal connective structure between its two most distinctive orbits.

7.2 Refinement of the dihedral orbit partition

The dihedral orbit partition of the fourteen-mode multiplet is:

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}. \quad (32)$$

The Klein orbit partition refines the dihedral orbit partition:

- $\mathcal{O}_0$ is the single Klein orbit $V_1 = \{\nu, e\}$.
- $\mathcal{O}_1$ decomposes into three Klein orbits: $V_3 = \{d, b\}$, $V_4 = \{\mu, c\}$, and $V_6 = \{\text{dark}, s, u, \tau\}$.
- $\mathcal{O}_2$ decomposes into two Klein orbits: $V_2 = \{H, Z\}$ and $V_5 = \{W, t\}$.

The Klein group therefore provides a fine-grained partition of the multiplet, in which each dihedral orbit of size greater than two is further decomposed.

7.3 Relation to the observer pair

The observer pair $\{W, t\}$ of HCSM-26 is the Klein orbit V_5 . Its pointwise stabilizer is $\{e, rs\}$ — the subgroup generated by the chiral reflection. This is consistent with the Chirality Selection Theorem: the observer pair is the unique fixed locus of the chiral reflection on the fourteen-mode multiplet.

The four-cycle orbit V_6 has trivial pointwise stabilizer. Its swap partner with the observer orbit is the top quark, which lies in V_5 . The two most distinctive orbits of the Klein structure are connected by the swap pair $\{\text{dark}, t\}$.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at the verification folder with the number twenty-seven. The script constructs the fourteen-mode multiplet, the diagonal reflection rs , the half-period translation T_{half} , the Klein four-group $V = \langle rs, T_{\text{half}} \rangle$, the orbit decomposition, the pointwise stabilizers, the distinguished-pair graph, and the four-cycle orbit's mass-ordering values from HCSM-native definitions, and verifies:

1. The Klein four-group presentation $rs^2 = T_{\text{half}}^2 = e$ and $[rs, T_{\text{half}}] = 0$.
2. The orbit decomposition into four two-cycle orbits with stabilizer $\{e, rs \cdot T_{\text{half}}\}$, one two-cycle orbit with stabilizer $\{e, rs\}$, and one four-cycle orbit with trivial stabilizer.
3. The pointwise stabilizer classification of each orbit.
4. The unique inter-orbit edge $\{\text{dark}, t\}$ in the distinguished-pair graph.
5. The four-cycle orbit's mass-ordering values and their framework-constant forms.
6. The additive identities $N/4(\tau) + N/4(\text{dark}) = N/4(u)$ and $N/4(\tau) + N/4(s) = H + 2d - 1$.
7. The sum = 56 and the mean = 14.

Running the master verification script reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has characterized the Klein orbit structure of the fourteen-mode multiplet in the Hodge Complex Standard Model. The Klein four-group generated by the diagonal reflection and the half-period translation acts on the multiplet with orbit decomposition consisting of five two-cycle orbits and one four-cycle orbit. The pointwise stabilizer of each orbit is one of the three non-trivial subgroups of the Klein group: four orbits have stabilizer $\{e, rs \cdot T_{\text{half}}\}$, the observer orbit has stabilizer $\{e, rs\}$, and the four-cycle orbit has trivial stabilizer. The four-cycle orbit contains the framework-constant shadow of the Hodge sector: its four mass-ordering labels are all framework constants, they satisfy two additive identities within the orbit, and their sum equals fifty-six, the dimension of the Hodge sector at the self-paired eigenvalue. Their mean equals fourteen, the multiplicity.

The orbit-class swap pair, consisting of the dark photon and the top quark, is the unique edge in the distinguished-pair graph that connects two distinct Klein orbits. It bridges the four-cycle orbit and the observer orbit, providing the framework's minimal connective structure between its two most distinctive orbits.

The Klein structure organizes the fourteen-mode multiplet in a way that refines the dihedral orbit partition and brings the framework-constant arithmetic of the four-cycle orbit into focus.

The observer pair of HCSM-26 is the orbit V_5 of this structure, and the framework-constant orbit is V_6 .

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-13: The Processor T and the F/B -Class Correspondence*, HCSM White Paper Series (2026).
- [3] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-23: The δ Function and $N/4$ Labels*, HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-26: The Observer Pair*, HCSM White Paper Series (2026).

A Web Crawler Metadata

A.1 Dublin Core

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DC.title: HCSM-27: Klein Orbit Structure
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; Klein four-group; dihedral
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DC.description: This paper characterizes the Klein orbit structure of
                the fourteen-mode multiplet in the Hodge Complex Standard Model. The
                Klein four-group generated by the diagonal reflection and the
                half-period translation acts on the multiplet with orbit decomposition
                consisting of five two-cycle orbits and one four-cycle orbit. The
                pointwise stabilizer of each orbit falls into one of three classes.
                The four-cycle orbit contains the framework-constant shadow of the
                Hodge sector: its four mass-ordering labels are all framework
                constants, they satisfy two additive identities, and their sum equals
                fifty-six, the dimension of the Hodge sector at the self-paired
                eigenvalue. The orbit-class swap pair is the unique inter-orbit edge
                in the distinguished-pair graph.
DC.identifier: HCSM-27-2026-09

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A.2 Schema.org JSON-LD

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    decomposition consisting of five two-cycle orbits and one four-cycle
    orbit. The pointwise stabilizer of each orbit falls into one of
    three classes. The four-cycle orbit contains the framework-constant
    shadow of the Hodge sector: its four mass-ordering labels are all
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    their sum equals fifty-six, the dimension of the Hodge sector at
    the self-paired eigenvalue. The orbit-class swap pair is the unique
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A.3 BibTeX for Automated Indexing

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  journal = {HCSM White Paper Series},
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  month = {September},
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    decomposition consisting of five two-cycle orbits and one four-cycle
    orbit. The four-cycle orbit's four mass-ordering labels are all
    framework constants, with sum fifty-six, the dimension of the Hodge
    sector at the self-paired eigenvalue.}
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